

Thyroid Hormones Explained

How a small gland affects energy, metabolism and daily function.

The control pathway

- The pituitary gland releases TSH, which signals the thyroid.
- The thyroid mainly produces T4 and a smaller amount of T3.
- T3 is the more biologically active hormone at many tissues.

When levels are low

- Hypothyroidism may cause fatigue, cold intolerance, constipation, dry skin, slowed thinking and weight gain.
- A common laboratory pattern is high TSH with low free T4 in primary hypothyroidism.

When levels are high

- Hyperthyroidism may cause palpitations, tremor, heat intolerance, anxiety, sweating and weight loss.
- A common laboratory pattern is low TSH with elevated thyroid hormones.

Common tests

- TSH and free T4 are central tests.
- Free T3 and thyroid antibodies may be used in selected situations.
- Ultrasound evaluates thyroid structure rather than hormone production.

Urgent warning

Persistent symptoms require medical assessment. Do not change thyroid medication without professional guidance.